Mid-term Exam (Signals and Systems - Fall 2026)

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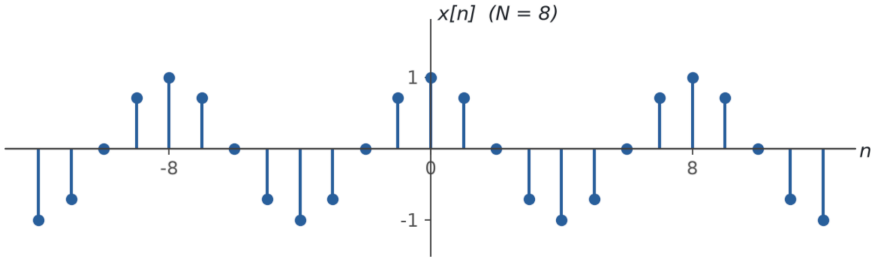
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1. (1장 Signals and Systems) Example 1. 그림의 $x[n]$ 이 몇 칸마다 되풀이되는지 보고 Fundamental Period N_0 를 찾는 문제예요.

Periodic signals — Example 1 (Discrete-time)

- A discrete-time periodic signal: the pattern repeats every ***N*** samples



- smallest such N = **fundamental period N_0**

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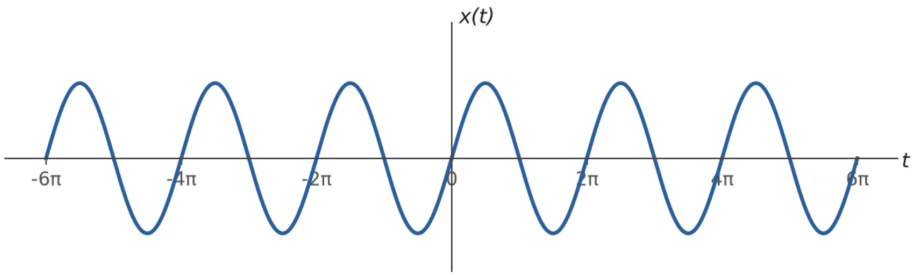
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2. (1장 Signals and Systems) Example 2. $x(t) = \sin(t)$ 가 얼마마다 되풀이되는지, Fundamental Period T_0 를 구하는 문제예요.

1.2 Transformations of the Independent Variable

Periodic signals — Example 2 (Continous-Time)

- $x(t) = sin(t)$ repeats with fundamental period $T_o = 2\pi$:



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3. (1장 Signals and Systems) Example 1.5. $x(t) = e^{j2t} + e^{j3t}$ 의 크기 $|x(t)|$ 를 구하는 문제예요. Euler's formula 를 쓰는 자리예요.

Continuous-Time Complex Exponential Signals

- **Periodic Complex Exponential and Sinusoidal signals**

- Example $x(t) = e^{j2t} + e^{j3t}.$

Euler's form

$$x(t) = e^{j2.5t}(e^{-j0.5t} + e^{j0.5t}),$$

$$x(t) = 2e^{j2.5t}\cos(0.5t).$$

Magnitude

$$|x(t)| = 2|\cos(0.5t)|.$$

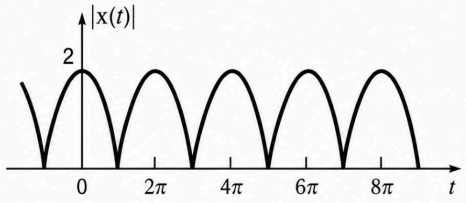


Figure 1.22 The full-wave rectified sinusoid of Example 1.5.

$$e^{j\theta} = \cos\theta + j\sin\theta \begin{cases} \cos\theta = \dfrac{e^{j\theta} + e^{-j\theta}}{2} \\ \sin\theta = \dfrac{e^{j\theta} - e^{-j\theta}}{2j} \end{cases}$$

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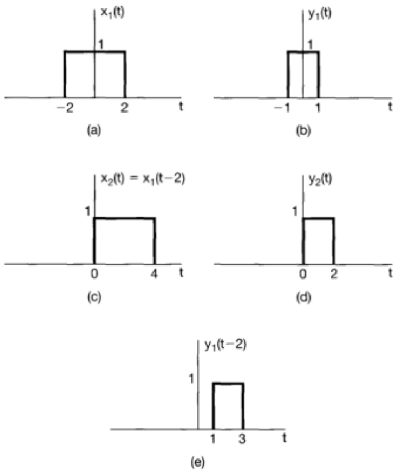
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4. (1장 Signals and Systems) Example . $y(t) = x(2t)$ 가 Time Invariant 인지 판정하는 문제예요. 강의자료가 그림 (a)~(e) 로 보여 줘요.

- **Time Invariance**
  - **Example**
  - Time scaling system is invariant or not?

$$y(t) = x(2t).$$



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5. (1장 Signals and Systems) Example 1.19. $y[n] = \mathrm{Re}\{x[n]\}$ 가 Linear 인지 판정하는 문제예요. 강의자료에 "Examle" 로 오타가 나 있어요.

Examle 1.19 | is y[n] = Re{x[n]} linear?

- subtle! additivity holds, and REAL scaling holds — but linearity allows **complex** constants:

$$y[n] = \mathrm{Re}\{x[n]\}$$

$$x_2[n] = j\,x_1[n] \Rightarrow y_2[n] = \mathrm{Re}\{jr[n] - s[n]\} = -s[n] \neq j\,y_1[n] = j\,r[n]$$

- scaling by j fails → **NOT linear** (over the complex numbers)

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6. (1장 Signals and Systems) Example 1.20. $y[n] = 2x[n] + 3$ 이 Linear 인지, 그리고 왜 Incrementally Linear 라고 부르는지 보는 문제예요.

- **Linearity**
  - **Example 1.20**

$x[n] \rightarrow 2x[n]$, Linear system

$y[n] = 2x[n] + 3.$ Superposition of two system

Incrementally linear system

$y_0[n] = 3.$ Zero-input response

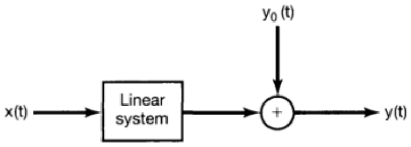


Figure 1.48 Structure of an incrementally linear system. Here, $y_0[n]$ is the zero-input response of the system.

$y_1[n] - y_2[n] = 2x_1[n] + 3 - \{2x_2[n] + 3\} = 2\{x_1[n] - x_2[n]\}$

The difference between responses of two inputs is linear.

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7. (1장 Signals and Systems) Homework 1.21. Figure P1.21 의 $x(t)$ 로 (a)~(f) 여섯 개를 그리는 문제예요. 이동, 반전, 척도가 한꺼번에 나와요.

- **Homework**

1.21. A continuous-time signal $x(t)$ is shown in Figure P1.21. Sketch and label carefully each of the following signals:

(a) $x(t-1)$ **(b)** $x(2-t)$ **(c)** $x(2t+1)$
(d) $x(4-\frac{t}{2})$ **(e)** $[x(t)+x(-t)]u(t)$ **(f)** $x(t)[\delta(t+\frac{3}{2})-\delta(t-\frac{3}{2})]$

1.22. A discrete-time signal is shown in Figure P1.22. Sketch and label carefully each of the following signals:

(a) $x[n-4]$ **(b)** $x[3-n]$ **(c)** $x[3n]$
(d) $x[3n+1]$ **(e)** $x[n]u[3-n]$ **(f)** $x[n-2]\delta[n-2]$
(g) $\frac{1}{2}x[n]+\frac{1}{2}(-1)^n x[n]$ **(h)** $x[(n-1)^2]$

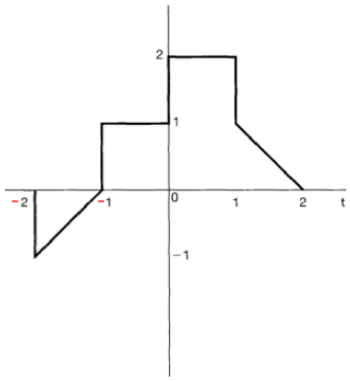


Figure P1.21

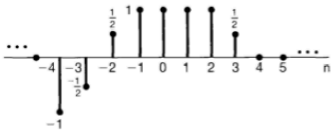


Figure P1.22

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8. (1장 Signals and Systems) Homework 1.22. Figure P1.22 의 $x[n]$ 으로 (a)~(h) 여덟 개를 그리는 문제예요. 칸이 사라지는 자리가 핵심이에요.

- **Homework**

1.21. A continuous-time signal $x(t)$ is shown in Figure P1.21. Sketch and label carefully each of the following signals:

(a) $x(t-1)$ **(b)** $x(2-t)$ **(c)** $x(2t+1)$

(d) $x(4-\frac{t}{2})$ **(e)** $[x(t)+x(-t)]u(t)$ **(f)** $x(t)[\delta(t+\frac{3}{2})-\delta(t-\frac{3}{2})]$

1.22. A discrete-time signal is shown in Figure P1.22. Sketch and label carefully each of the following signals:

(a) $x[n-4]$ **(b)** $x[3-n]$ **(c)** $x[3n]$

(d) $x[3n+1]$ **(e)** $x[n]u[3-n]$ **(f)** $x[n-2]\delta[n-2]$

(g) $\frac{1}{2}x[n]+\frac{1}{2}(-1)^n x[n]$ **(h)** $x[(n-1)^2]$

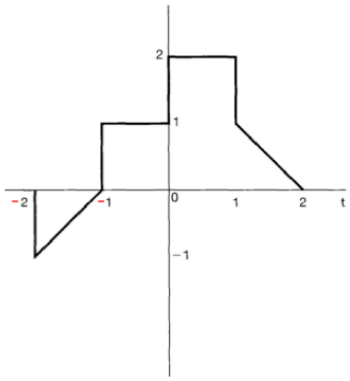


Figure P1.21

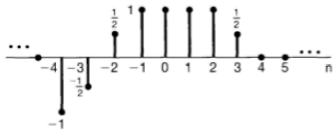


Figure P1.22

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9. (1장 Signals and Systems) Homework 1.25. 연속시간 신호가 주기인지 보고, 주기라면 Fundamental Period 를 구하는 문제예요. 강의자료에 교수님이 빨간 X 로 뺀 항목이 있어요.

- **Homework**

1.25. Determine whether or not each of the following continuous-time signals is periodic. If the signal is periodic, determine its fundamental period.

(a) $x(t) = 3\cos(4t + \frac{\pi}{3})$ **(b)** $x(t) = e^{j(\pi t - 1)}$

(c) $x(t) = [\cos(2t - \frac{\pi}{3})]^2$ **(d)** $x(t) = \mathcal{E}v\{\cos(4\pi t)u(t)\}$

(e) $x(t) = \mathcal{E}v\{\sin(4\pi t)u(t)\}$ **(f)** $x(t) = \sum_{n=-\infty}^{\infty} e^{-(2t-n)}u(2t-n)$

- hints: (a) $T = 2\pi/4$ · (b) complex exponential — always periodic in CT · (c) square the cosine first
- use this week's tools: $\omega_0 \rightarrow T$ $2\pi/\omega_0$, and check the smallest common period

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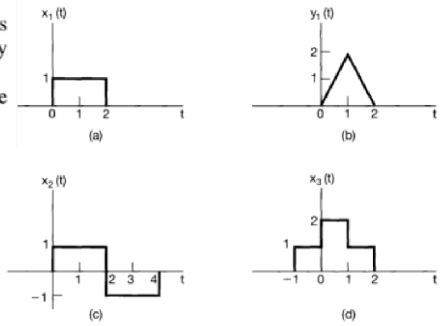
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10. (1장 Signals and Systems) Homework 1.31. x_1 의 응답 y_1 을 알 때 x_2, x_3 의 응답을 구하는 문제예요. 이 과목 전체에서 가장 중요한 성질을 쓰는 자리예요.

1.31. In this problem, we illustrate one of the most important consequences of the properties of linearity and time invariance. Specifically, once we know the response of a linear system or a linear time-invariant (LTI) system to a single input or the responses to several inputs, we can directly compute the responses to many other input signals. Much of the remainder of this book deals with a thorough exploitation of this fact in order to develop results and techniques for analyzing and synthesizing LTI systems.

(a) Consider an LTI system whose response to the signal $x_1(t)$ in Figure P1.31(a) is the signal $y_1(t)$ illustrated in Figure P1.31(b). Determine and sketch carefully the response of the system to the input $x_2(t)$ depicted in Figure P1.31(c).

(b) Determine and sketch the response of the system considered in part (a) to the input $x_3(t)$ shown in Figure P1.31(d).



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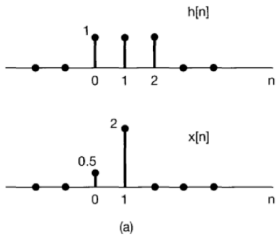
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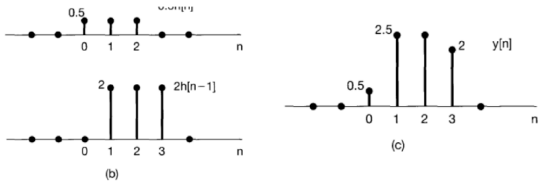
11. (2장 LTI Systems) Example 2.1. $h[n]$ = [0 0 1 1 1 0 0], $x[n]$ = [0 0 0.5 2 0 0 0] 일 때 $y[n]$ 을 구하는 문제 예요.

- **The Discrete-Time Unit Impulse Response and the Convolution Sum Representation of LTI Systems**
  - **Example 2.1**
    - Consider a LTI system with the following unit impulse response:
      $h[n] = [0\ 0\ 1\ 1\ 1\ 0\ 0]$
    - For the input sequence:
      $x[n] = [0\ 0\ 0.5\ 2\ 0\ 0\ 0]$
    - The result is: $y[n] = x[0]h[n-0] + x[1]h[n-1] = 0.5h[n] + 2h[n-1]$



(a)

$y[n]$ = … + x[0]h[n] + x[1]h[n-1] + …
= 0 +
0.5*[0 0 1 1 1 0 0] +
2.0*[0 0 0 1 1 1 0] +
0
= [0 0 0.5 2.5 2.5 2 0]



(b) (c)

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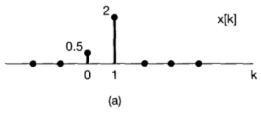
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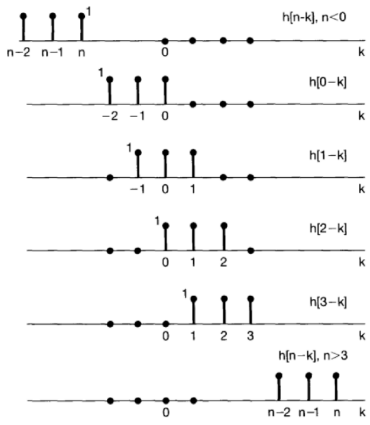
12. (2장 LTI Systems) Example 2.2. 같은 x, h 를 이번에는 그래프 방법으로 푸는 문제예요. $h[n-k]$ 를 그려 보는 것이 핵심이에요.

- **The Discrete-Time Unit Impulse Response and the Convolution Sum Representation of LTI Systems**
  - **Example 2.2**
    - Sketch $x[k]$ and $h[n-k]$ for any particular value of n, then multiply the two signals and sum over all values of k.
    - For $n < 0$, we see that $x[k]h[n-k] = 0$ for all k, since the non-zero values of the two signals do not overlap.

$y[0] = \Sigma_k x[k]h[0-k] = 0.5$
$y[1] = \Sigma_k x[k]h[1-k] = 0.5+2$
$y[2] = \Sigma_k x[k]h[2-k] = 0.5+2$
$y[3] = \Sigma_k x[k]h[3-k] = 2$



(a)



(b)

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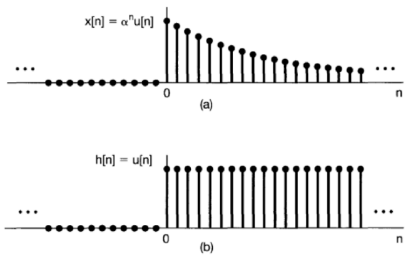
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13. (2장 LTI Systems) Example 2.3. $h[n] = u[n]$ 인 시스템에 $x[n] = \alpha^n u[n]$ $(0 < \alpha < 1)$ 을 넣었을 때 출력을 구하는 문제예요.

2.1 Discrete-time LTI Systems: The Convolution Sum

- **The Discrete-Time Unit Impulse Response and the Convolution Sum Representation of LTI Systems**
  - **Example 2.3**
    - Consider a LTI system that has a step response $h[n] = u[n]$ to the unit impulse input signal
    - What is the response when an input signal of the form $x[n] = \alpha^n u[n]$ where $0 < \alpha < 1$, is applied?

$$x[k]h[n-k] = \begin{cases} \alpha^k, & 0 \le k \le n \\ 0, & \text{otherwise} \end{cases}$$



S3 p.29

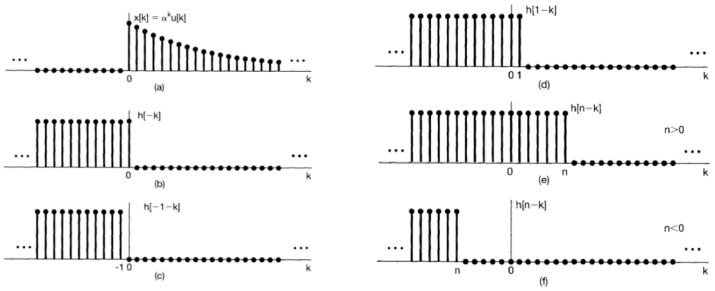
2.1 Discrete-time LTI Systems: The Convolution Sum

- **The Discrete-Time Unit Impulse Response and the Convolution Sum Representation of LTI Systems**
  - **Example 2.3** $x[n] = \alpha^n u[n]$
    - For $n \ge 0$:

$$y[n] = \sum_{k=0}^{n} \alpha^k$$

$$= \frac{1-\alpha^{n+1}}{1-\alpha}$$

$$y[n] = \left(\frac{1-\alpha^{n+1}}{1-\alpha}\right) u[n]$$



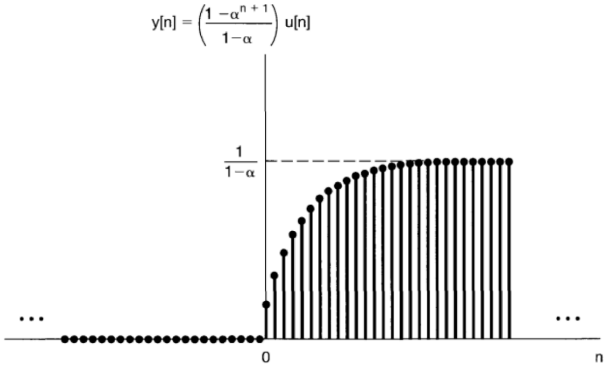
S3 p.30

- **The Discrete-Time Unit Impulse Response and the Convolution Sum Representation of LTI Systems**
  - **Example 2.3** $x[n] = \alpha^n\, u[n]$
    - For $n \geq 0$:

$$y[n] = \sum_{k=0}^{n} \alpha^k$$

$$= \frac{1-\alpha^{n+1}}{1-\alpha}$$

$$y[n] = \left(\frac{1-\alpha^{n+1}}{1-\alpha}\right) u[n]$$



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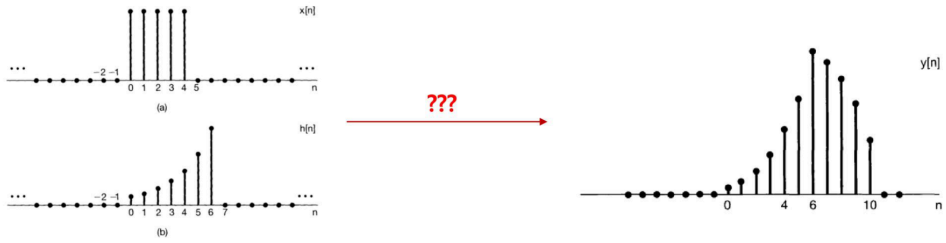
Note: Students should write in English for evaluation.

14. (2장 LTI Systems) Example 2.4. $x[n] = 1$ $(0 \le n \le 4)$, $h[n] = \alpha^n$ $(0 \le n \le 6)$ 의 Convolution 이에요. 둘 다 길이가 유한해요.

- **The Discrete-Time Unit Impulse Response and the Convolution Sum Representation of LTI Systems**
  - **Example 2.4**

$$x[n] = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & \text{otherwise} \end{cases}$$

$$h[n] = \begin{cases} \alpha^n, & 0 \le n \le 6 \\ 0, & \text{otherwise} \end{cases}$$



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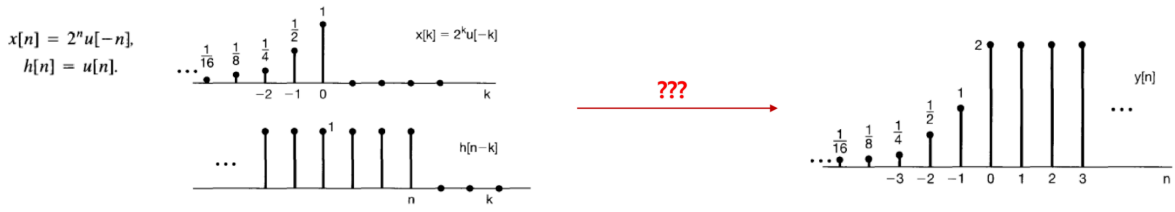
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15. (2장 LTI Systems) Example 2.5. $x[n] = 2^n u[-n]$, $h[n] = u[n]$ 의 Convolution 이에요. $u[-n]$ 이라 신호가 왼쪽으로 뻗어 있어요.

- **The Discrete-Time Unit Impulse Response and the Convolution Sum Representation of LTI Systems**
  - **Example 2.5**



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16. (2장 LTI Systems) Example 2.6. $x(t) = e^{-at}u(t)$ (a>0), $h(t) = u(t)$ 의 Convolution 적분이에요. 연속시간 쪽의 기본 문제예요.

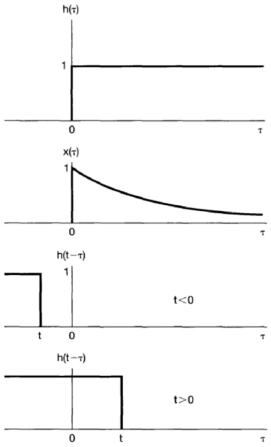
2.2 Continuous-time LTI Systems: The Convolution Integral

- **Continuous-Time Unit Impulse Response and the Convolution Integral Representation of LTI Systems**
  - **Example 2.6.**
    - Let $x(t)$ be the input to a LTI system with unit impulse response $h(t)$:

$$x(t) = e^{-at}u(t) \quad a > 0$$
$$h(t) = u(t)$$

    - For $t > 0$:

$$x(\tau)h(t-\tau) = \begin{cases} e^{-a\tau} & 0 < \tau < t \\ 0 & \text{otherwise} \end{cases}$$



S3 p.45

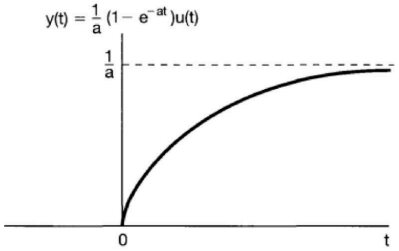
2.2 Continuous-time LTI Systems: The Convolution Integral

- **Continuous-Time Unit Impulse Response and the Convolution Integral Representation of LTI Systems**
  - **Example 2.6.**
    - We can compute $y(t)$ for $t > 0$:

$$y(t) = \int_0^t e^{-a\tau}d\tau = -\frac{1}{a}e^{-a\tau}\Big|_0^t$$
$$= \frac{1}{a}\left(1 - e^{-at}\right)$$

    - So for all t:

$$y(t) = \frac{1}{a}(1 - e^{-at})u(t)$$



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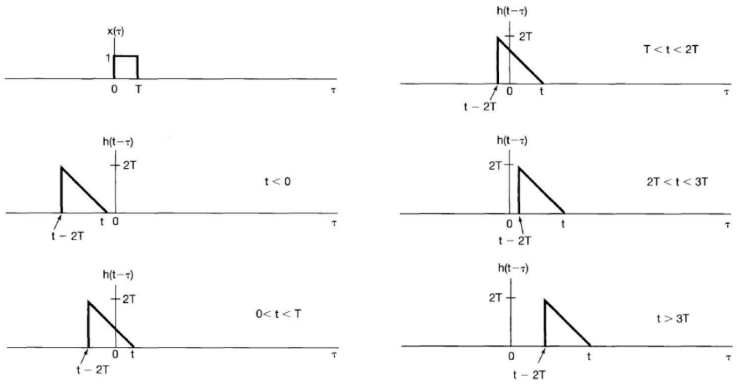
17. (2장 LTI Systems) Example 2.7. $x(t) = 1$ ($0<t<T$), $h(t) = t$ ($0<t<2T$) 의 Convolution 이에요. 구간이 가장 많이 갈리는 문제예요.

- **Continuous-Time Unit Impulse Response and the Convolution Integral Representation of LTI Systems**
  - **Example 2.7.**

$$x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$

$$h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & \text{otherwise} \end{cases}$$

$$y(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{2}t^2, & 0 < t < T \\ Tt - \frac{1}{2}T^2, & T < t < 2T \\ -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2, & 2T < t < 3T \\ 0, & 3T < t \end{cases}$$



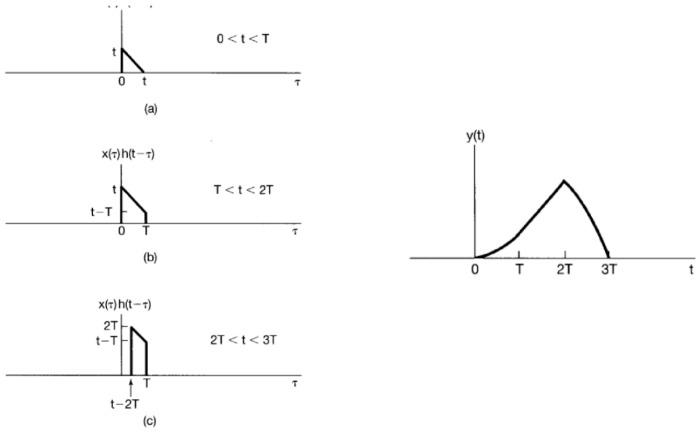
S3 p.47

- **Continuous-Time Unit Impulse Response and the Convolution Integral Representation of LTI Systems**
  - **Example 2.7.**

$$x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$

$$h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & \text{otherwise} \end{cases}$$

$$y(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{2}t^2, & 0 < t < T \\ Tt - \frac{1}{2}T^2, & T < t < 2T \\ -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2, & 2T < t < 3T \\ 0, & 3T < t \end{cases}$$



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18. (2장 LTI Systems) Example 2.8. $x(t) = e^{2t}u(-t)$, $h(t) = u(t-3)$ 의 Convolution 이에요. 둘 다 방향이 꼬여 있어 헷갈리기 쉬워요.

2.2 Continuous-time LTI Systems: The Convolution Integral

- **Continuous-Time Unit Impulse Response and the Convolution Integral Representation of LTI Systems**
  - **Example 2.8.**

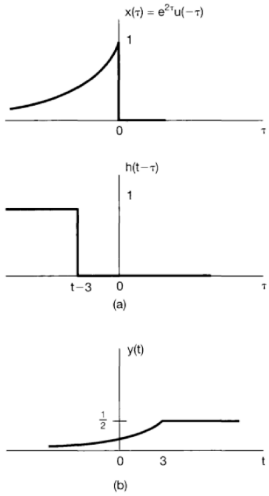
$$x(t) = e^{2t}u(-t)$$
$$h(t) = u(t-3)$$

↓

$$y(t) = \int_{-\infty}^{t-3} e^{2\tau}d\tau = \frac{1}{2}e^{2(t-3)}$$

For $t-3 \geq 0$, the product $x(\tau)h(t-\tau)$ is non-zero for $-\infty < \tau < 0$, so the convolution integral becomes:

$$y(t) = \int_{-\infty}^{0} e^{2\tau}d\tau = \frac{1}{2}$$



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19. (2장 LTI Systems) Example 2.10. $x[n] = 0.5^n\,u[n] + 2^n\,u[-n]$, $h[n] = u[n]$ 이에요. x 가 모든 n 에서 0 이 아니라 그대로는 못 풀어요.

- **Distributive Property (Parallel Systems)**
  - **Example 2.10.**
  - Let $y[n]$ denote the convolution of the following two sequences:

$$x[n] = 0.5^n u[n] + 2^n u[-n]$$
$$h[n] = u[n]$$

  - $x[n]$ is non-zero for all n. We will use the **distributive** property to express $y[n]$ as the sum of two simpler convolution problems.

$$y[n] = (x_1[n] + x_2[n]) * h[n]$$

$x_1[n] = 0.5^n u[n]$, $x_2[n] = 2^n u[-n]$

$$y[n] = y_1[n] + y_2[n],$$

$y_1[n] = x_1[n] * h[n]$, $y_1[n] = x_1[n] * h[n]$

S3 p.56

- **Distributive Property (Parallel Systems)**
  - **Example 2.10.**

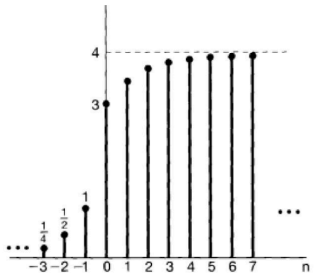
$$y[n] = y_1[n] + y_2[n],$$

$y_1[n] = x_1[n] * h[n]$, $y_1[n] = x_1[n] * h[n]$

$$y_1[n] = \left(\frac{1-0.5^{n+1}}{1-0.5}\right) u[n]$$

$$y_2[n] = \begin{cases} 2^{n+1} & n \le 0 \\ 2 & n \ge 1 \end{cases}$$

Example 2.3 & 2.5



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20. (2장 LTI Systems) Example 2.11. $h(t) = \delta(t - t_0)$ 인 시스템의 Inverse System 을 구하는 문제예요. Inverse System 의 가장 쉬운 예예요.

- **Invertibility of LTI Systems**
  - The aim is to calculate “inverse systems” such that
  - The resulting **serial** system is therefore **memoryless**
  - **Example 2.11.**

$$h(t) = \delta(t - t_0). \qquad h_1(t) = \delta(t + t_0),$$

$$x(t - t_0) = x(t) * \delta(t - t_0). \qquad h(t) * h_1(t) = \delta(t - t_0) * \delta(t + t_0) = \delta(t).$$

The convolution of a signal with a shifted impulse simply shifts the signal.

To recover the input from the output, i.e., to invert the system, all that is required is to shift the output back.

The system with this compensating time shift.

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21. (2장 LTI Systems) Example 2.12. $h[n] = u[n]$ 인 Accumulator 의 Inverse System 을 구하는 문제예요. 더한 것을 되돌리는 방법을 묻고 있어요.

2.3 Properties of Linear Time-Invariant Systems

- **Invertibility of LTI Systems**
  - **Example 2.12.**
  - Consider a DT LTI system with an impulse response $h[n] = u[n]$
  - Using convolution, the response to an arbitrary input $x[n]$:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

  - As $u[n-k] = 0$ for $n-k < 0$ and 1 for $n-k \geq 0$, this becomes

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$

  - i.e. it acts as a **running sum or accumulator**.

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2.3 Properties of Linear Time-Invariant Systems

- **Invertibility of LTI Systems**
  - **Example 2.12.**
  - Therefore an inverse system can be expressed as:

$$y[n] = x[n] - x[n-1]$$

  - A **first difference (differential) operator**, which has an impulse response

$$h_1[n] = \delta[n] - \delta[n-1]$$

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22. (2장 LTI Systems) Example 2.13. 시간 이동 시스템과 Integrator(Accumulator)가 Stable 한지 판정하는 문제예요. 판정 기준을 외워 쓰는 자리예요.

- **Stability for LTI Systems**
  - Therefore a **DT LTI system is stable** if and only if its **impulse response is absolutely summable**, ie.

$$\sum_{k=-\infty}^{\infty}|h[k]|<\infty \qquad \int_{-\infty}^{\infty}|h(\tau)|d\tau<\infty$$ CT system: absolutely integrable

  - **Example 2.13.**
  - Are the DT and CT pure time shift systems stable?

$$h[n]=\delta[n-n_0] \longrightarrow \sum_{k=-\infty}^{\infty}|h[k]|=\sum_{k=-\infty}^{\infty}|\delta[k-n_0]|=1<\infty$$
$$h(t)=\delta(t-t_0) \longrightarrow \int_{-\infty}^{\infty}|h(\tau)|d\tau=\int_{-\infty}^{\infty}|\delta(\tau-t_0)|d\tau=1<\infty$$

Therefore, both the CT and DT systems are **stable**: all finite input signals produce a finite output signal

S3 p.69

- **Stability for LTI Systems**
  - **Example 2.13.**
  - Are the discrete and continuous-time integrator systems stable?

$$h[n]=u[n-n_0] \longrightarrow \sum_{k=-\infty}^{\infty}|h[k]|=\sum_{k=-\infty}^{\infty}|u[k-n_0]|=\sum_{k=n_0}^{\infty}|u[k]|=\infty$$
$$h(t)=u(t-t_0) \longrightarrow \int_{-\infty}^{\infty}|h(\tau)|d\tau=\int_{-\infty}^{\infty}|u(\tau-t_0)|d\tau=\int_{t_0}^{\infty}|u(\tau)|d\tau=\infty$$

Therefore, both the CT and DT systems are **unstable**: at least one finite input causes an infinite output signal

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23. (2장 LTI Systems) Example 2.15. $y[n]$ - \tfrac12 $y[n$-1] = $x[n]$ 에 $x[n] = K\delta[n]$ 을 넣어 Impulse Response 를 구하는 문제예요.

- **Linear Constant-Coefficient Difference Equations**
  - **Example 2.15**

$$y[n] - \frac{1}{2}y[n-1] = x[n].$$

$$\downarrow$$

$$y[n] = x[n] + \frac{1}{2}y[n-1],$$

For the condition of initial rest:

$$x[n] = K\delta[n].$$

$$\downarrow$$

$$y[0] = x[0] + \frac{1}{2}y[-1] = K,$$

$$y[1] = x[1] + \frac{1}{2}y[0] = \frac{1}{2}K,$$

$$y[2] = x[2] + \frac{1}{2}y[1] = \left(\frac{1}{2}\right)^2 K, \longrightarrow h[n] = \left(\frac{1}{2}\right)^n u[n].$$

$$\vdots$$

$$y[n] = x[n] + \frac{1}{2}y[n-1] = \left(\frac{1}{2}\right)^n K.$$

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24. (2장 LTI Systems) Homework 2.21. $y[n] = x[n] * h[n]$ 을 (a)~(d) 네 쌍에 대해 구하는 문제예요. 손으로 하는 Convolution 연습 그 자체예요.

2.7 Causal LTI Systems Described by Differential and Difference Equations

- **Homework**

2.21. Compute the convolution $y[n] = x[n] * h[n]$ of the following pairs of signals:

(a) $\left.\begin{aligned} x[n] &= \alpha^n u[n], \\ h[n] &= \beta^n u[n], \end{aligned}\right\}$ $\alpha \neq \beta$

(b) $x[n] = h[n] = \alpha^n u[n]$

(c) $x[n] = (-\frac{1}{2})^n u[n-4]$
$h[n] = 4^n u[2-n]$

(d) $x[n]$ and $h[n]$ are as in Figure P2.21.

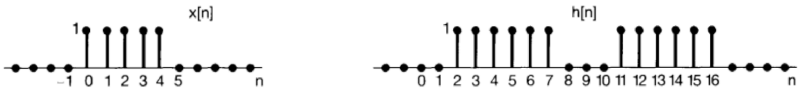


Figure P2.21

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25. (2장 LTI Systems) Homework 2.23. 삼각 펄스 $h(t)$ 와 Impulse 열 $x(t) = \Sigma_k \delta(t\text{-}kT)$ 의 Convolution 을 T 네 값에 대해 그리는 문제예요.

2.7 Causal LTI Systems Described by Differential and Difference Equations

- **Homework**

2.23. Let $h(t)$ be the triangular pulse shown in Figure P2.23(a), and let $x(t)$ be the impulse train depicted in Figure P2.23(b). That is,

$$x(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT).$$

Determine and sketch $y(t) = x(t) * h(t)$ for the following values of T:
(a) $T = 4$ **(b)** $T = 2$ **(c)** $T = 3/2$ **(d)** $T = 1$

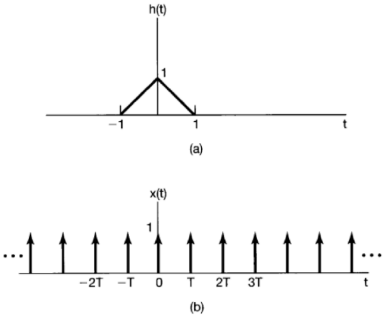


Figure P2.23

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26. (2장 LTI Systems) Homework 2.28. Impulse Response (a)~(g) 가 각각 Causal 인지 Stable 인지 판정하고 이유를 쓰는 문제예요. 시험에 그대로 나오기 좋은 모양이에요.

2.7 Causal LTI Systems Described by Differential and Difference Equations

- **Homework**

2.28. The following are the impulse responses of discrete-time LTI systems. Determine whether each system is causal and/or stable. Justify your answers.

(a) $h[n] = (\frac{1}{5})^n u[n]$

(b) $h[n] = (0.8)^n u[n+2]$

(c) $h[n] = (\frac{1}{2})^n u[-n]$

(d) $h[n] = (5)^n u[3-n]$

(e) $h[n] = (-\frac{1}{2})^n u[n] + (1.01)^n u[n-1]$

(f) $h[n] = (-\frac{1}{2})^n u[n] + (1.01)^n u[1-n]$

(g) $h[n] = n(\frac{1}{3})^n u[n-1]$

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27. (2장 LTI Systems) Homework 2.31. $y[n] + 2y[n\text{-}1] = x[n] + 2x[n\text{-}2]$ 를 Initial Rest 에서 재귀로 푸는 문제예요. 예제 2.15 와 같은 방법이에요.

2.1 Causal LTI Systems Described by Differential and Difference Equations

- **Homework**

2.31. Consider the LTI system initially at rest and described by the difference equation

$$y[n] + 2y[n-1] = x[n] + 2x[n-2].$$

Find the response of this system to the input depicted in Figure P2.31 by solving the difference equation recursively.

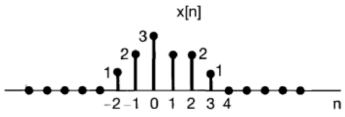


Figure P2.31

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Note: Students should write in English for evaluation.

28. (3장 Fourier Series) Example 3.1. $y(t) = x(t\text{-}3)$ 에 $x(t) = e^{j2t}$ 를 넣어 eigenvalue 를 구하고, 같은 시스템에 $x(t) = \cos 4t + \cos 7t$ 를 넣은 답까지 구하는 문제예요.

3.2 The Response of LTI Systems to Complex Exponentials

- **Eigenfunction**
  - **Example 3.1** $y(t) = x(t-3)$.

$$x(t) = e^{j2t}$$

$$y(t) = e^{j2(t-3)} = \underbrace{e^{-j6}}_{\text{eigenvalue}} \underbrace{e^{j2t}}_{\text{eigenfunction}}.$$

$$h(t) = \delta(t-3).$$

$$H(s) = \int_{-\infty}^{+\infty} \delta(\tau-3)e^{-s\tau}d\tau = e^{-3s},$$

$$H(j2) = e^{-j6}$$

$$x(t) = \cos(4t) + \cos(7t).$$

$$y(t) = \cos(4(t-3)) + \cos(7(t-3)).$$

$$x(t) = \frac{1}{2}e^{j4t} + \frac{1}{2}e^{-j4t} + \frac{1}{2}e^{j7t} + \frac{1}{2}e^{-j7t}.$$

$$y(t) = \frac{1}{2}e^{-j12}e^{j4t} + \frac{1}{2}e^{j12}e^{-j4t} + \frac{1}{2}e^{-j21}e^{j7t} + \frac{1}{2}e^{j21}e^{-j7t},$$

$$= \frac{1}{2}e^{j4(t-3)} + \frac{1}{2}e^{-j4(t-3)} + \frac{1}{2}e^{j7(t-3)} + \frac{1}{2}e^{-j7(t-3)}$$

$$= \cos(4(t-3)) + \cos(7(t-3)).$$

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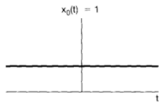
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29. (3장 Fourier Series) Example 3.2. a_k 일곱 개를 주고 $x(t) = \Sigma_{k=-3}^{+3}\, a_k\, e^{jk2\pi\, t}$ 를 \cos 들의 합으로 다시 쓰는 문제예요.

- **Linear Combinations of Harmonically Related Complex Exponentials**
  - **Example 3.2.**

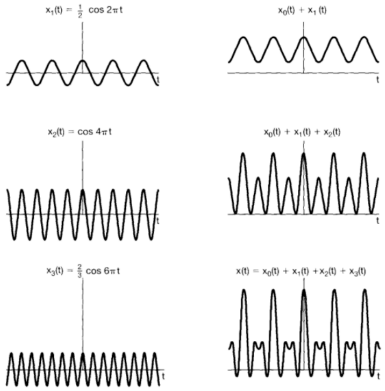


$$x(t) = \sum_{k=-3}^{+3} a_k e^{jk2\pi t},$$

$$a_0 = 1, \quad a_1 = a_{-1} = \frac{1}{4}, \quad a_2 = a_{-2} = \frac{1}{2}, \quad a_3 = a_{-3} = \frac{1}{3}.$$

$$\rightarrow \quad x(t) = 1 + \frac{1}{4}(e^{j2\pi t} + e^{-j2\pi t}) + \frac{1}{2}(e^{j4\pi t} + e^{-j4\pi t}) + \frac{1}{3}(e^{j6\pi t} + e^{-j6\pi t}).$$

$$x(t) = 1 + \frac{1}{2}\cos 2\pi t + \cos 4\pi t + \frac{2}{3}\cos 6\pi t.$$



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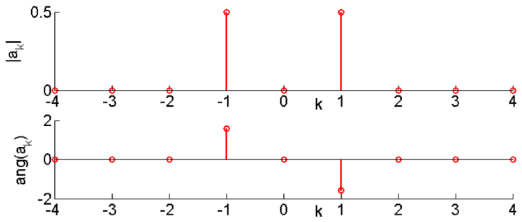
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30. (3장 Fourier Series) Example 3.3. $x(t) = \sin(\omega_0 t)$ 의 a_k 를 눈으로 읽는 방법과 적분으로 구하는 방법, 두 가지로 푸는 문제예요.

- **Determination of the Fourier Series Representation of a Continuous-time Periodic Signal**
  - **Example 3.3.**
    - The fundamental period of $x(t) = \sin(\omega_0 t)$ is ω_0
    - By inspection we can write: $\sin(\omega_0 t) = \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t}$
    - So $a_1 = 1/2j$, $a_{-1} = -1/2j$ and $a_k = 0$ otherwise
    - The magnitude and angle of the Fourier coefficients are:



S5 p.30

- **Determination of the Fourier Series Representation of a Continuous-time Periodic Signal**
  - **Example 3.3.**
    - The Fourier coefficients can also be explicitly evaluated $a_n = \frac{1}{T}\int_0^T x(t)e^{-jn\omega_0 t}dt$

$$a_0 = \frac{\omega_0}{2\pi}\int_0^{2\pi/\omega_0} \sin(\omega_0 t)dt = -\cos(\omega_0 t)\Big|_0^{2\pi/\omega_0} = 1 - 1 = 0$$

$$a_k = \frac{\omega_0}{2\pi}\int_0^{2\pi/\omega_0} \sin(\omega_0 t)e^{-jk\omega_0 t}dt = \frac{\omega_0}{2\pi}\int_0^{2\pi/\omega_0} \left(\frac{1}{2j}e^{j\omega_0 t} - \frac{1}{2j}e^{-j\omega_0 t}\right)e^{-jk\omega_0 t}dt$$

$$= \frac{\omega_0}{2\pi}\int_0^{2\pi/\omega_0} \left(\frac{1}{2j}e^{j\omega_0 t} - \frac{1}{2j}e^{-j\omega_0 t}\right)e^{-jk\omega_0 t}dt$$

$$= \frac{\omega_0}{4\pi j}\int_0^{2\pi/\omega_0} e^{-j(k-1)\omega_0 t} - e^{-j(k+1)\omega_0 t}dt$$

When k = +1 or −1, the integrals evaluate to T and $-T$, respectively. Otherwise the coefficients are zero. Therefore $a_1 = 1/2j$, $a_{-1} = -1/2j$

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Note: Students should write in English for evaluation.

31. (3장 Fourier Series) Example 3.4. $x(t) = 1 + \sin\omega_0 t + 2\cos\omega_0 t + \cos(2\omega_0 t + \pi/4)$ 의 a_k 를 모두 구하는 문제예요.

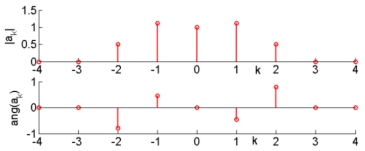
- **Determination of the Fourier Series Representation of a Continuous-time Periodic Signal**
  - **Example 3.4.**
    - Consider the additive sinusoidal series which has a fundamental frequency ω_0:

$$x(t) = 1 + \sin\omega_0 t + 2\cos\omega_0 t + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

$$x(t) = 1 + \frac{1}{2j}(e^{j\omega_0 t} - e^{-j\omega_0 t}) + (e^{j\omega_0 t} + e^{-j\omega_0 t}) + \frac{1}{2}(e^{j(2\omega_0 t + \pi/4)} + e^{-j(2\omega_0 t + \pi/4)})$$

$$= 1 + \left(1 + \frac{1}{2j}\right)e^{j\omega_0 t} + \left(1 - \frac{1}{2j}\right)e^{-j\omega_0 t} + \frac{1}{2}e^{j\pi/4}e^{j2\omega_0 t} + \frac{1}{2}e^{-j\pi/4}e^{-j2\omega_0 t}$$

$$a_0 = 1 \quad a_1 = \left(1 - \frac{1}{2}j\right) \quad a_{-1} = \left(1 + \frac{1}{2}j\right) \quad a_2 = \frac{1}{2}e^{j\pi/4} \quad a_{-2} = \frac{1}{2}e^{-j\pi/4}$$



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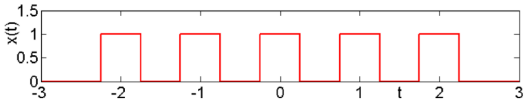
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Note: Students should write in English for evaluation.

32. (3장 Fourier Series) Example 3.5. $|t| < T_1$ 에서 1, $T_1 < |t| < T/2$ 에서 0 인 주기 사각파의 a_k 를 구하고, $T = 4T_1$ 일 때 값을 적는 문제예요.

- **Determination of the Fourier Series Representation of a Continuous-time Periodic Signal**
  - **Example 3.5.**
    - Consider the periodic square wave

$$x(t) = \begin{cases} 1 & |t| < T_1 \\ 0 & T_1 < |t| < T/2 \end{cases}$$



    - Fourier Coefficients:

$$a_0 = \frac{1}{T}\int_{-T_1}^{T_1} 1dt = \frac{2T_1}{T}$$

$$\begin{aligned} a_k &= \frac{1}{T}\int_{-T_1}^{T_1} e^{-jk\omega_0 t}dt = -\frac{1}{jk\omega_0 T}e^{-jk\omega_0 t}\Big|_{-T_1}^{T_1} \\ &= \frac{2}{k\omega_0 T}\left(\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{2j}\right) \\ &= 2\sin(k\omega_0 T_1)/k\omega_0 T \\ &= \sin(k\omega_0 T_1)/k\pi \end{aligned}$$

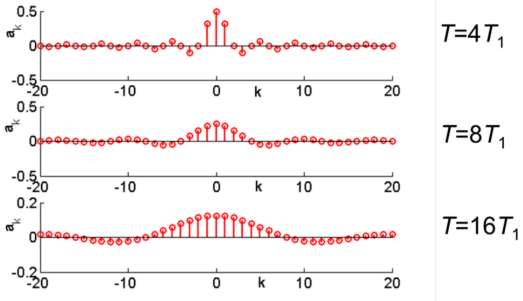
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- **Determination of the Fourier Series Representation of a Continuous-time Periodic Signal**
  - **Example 3.5.**
    - Because these coefficients are real, we only need to plot the value of the coefficients.
    - Note we have an infinite series of non-zero coefficients

For example, if $T = 4T_1$

$$a_0 = \frac{1}{T}\int_{-T_1}^{T_1} 1dt = \frac{2T_1}{T}$$

$$\begin{aligned} a_k &= \frac{1}{T}\int_{-T_1}^{T_1} e^{-jk\omega_0 t}dt = -\frac{1}{jk\omega_0 T}e^{-jk\omega_0 t}\Big|_{-T_1}^{T_1} \\ &= \frac{2}{k\omega_0 T}\left(\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{2j}\right) \\ &= 2\sin(k\omega_0 T_1)/k\omega_0 T \\ &= \sin(k\omega_0 T_1)/k\pi \end{aligned}$$



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33. (3장 Fourier Series) Homework 3.22. 주기가 주어진 두 신호의 Fourier 급수를 구하는 문제예요. (b) 는 주기 2 에 $x(t) = e^{-t}$, (c) 는 주기 4 에 반쪽만 \sin π t 예요.

- **Homework**

3.22. Determine the Fourier series representations for the following signals:

(b) $x(t)$ periodic with period 2 and

$$x(t) = e^{-t} \quad \text{for} \quad -1 < t < 1$$

(c) $x(t)$ periodic with period 4 and

$$x(t) = \begin{cases} \sin \pi t, & 0 \le t \le 2 \\ 0, & 2 < t \le 4 \end{cases}$$

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34. (3장 Fourier Series) Homework 3.24. 삼각파 $x(t)$ 가 주기 2 로 주어져요. a_0 를 구하고, dx/dt 의 Fourier 급수를 구한 뒤, 미분 성질 로 되짚어 $x(t)$ 의 계수를 구하는 문제예요.

- **Homework**

3.24. Let

$$x(t) = \begin{cases} t, & 0 \le t \le 1 \\ 2 - t, & 1 \le t \le 2 \end{cases}$$

be a periodic signal with fundamental period $T = 2$ and Fourier coefficients a_k.

(a) Determine the value of a_0.

(b) Determine the Fourier series representation of $dx(t)/dt$.

(c) Use the result of part (b) and the differentiation property of the continuous-time Fourier series to help determine the Fourier series coefficients of $x(t)$.

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35. (3장 Fourier Series) Homework 3.25. $x(t) = \cos 4\pi t$, $y(t) = \sin 4\pi t$, $z(t) = x(t)y(t)$ 의 계수를 구해요. (c) 는 곱셈 성질 로, (d) 는 직접 펼쳐서 구하고 둘을 비교해요.

- **Homework**

3.25. Consider the following three continuous-time signals with a fundamental period of $T = 1/2$:

$$x(t) = \cos(4\pi t),$$
$$y(t) = \sin(4\pi t),$$
$$z(t) = x(t)y(t).$$

(a) Determine the Fourier series coefficients of $x(t)$.
(b) Determine the Fourier series coefficients of $y(t)$.
(c) Use the results of parts (a) and (b), along with the multiplication property of the continuous-time Fourier series, to determine the Fourier series coefficients of $z(t) = x(t)y(t)$.
(d) Determine the Fourier series coefficients of $z(t)$ through direct expansion of $z(t)$ in trigonometric form, and compare your result with that of part (c).